Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Parameter updates can reinforce or cancel one another even when training loss continues to decrease. An exact finite-time decomposition of stochastic gradient descent (SGD) displacement separates gradient magnitudes from interactions across all temporal lags. Consecutive-gradient cosine provides a local directional diagnostic, while normalized displacement measures cumulative retention of traveled distance. Local expansions connect these quantities to curvature and sampling noise without identifying alignment sign with stability or objective progress. Full-batch experiments exhibit gradient reversal alongside continued loss reduction. Matched mini-batch continuations show that sampling changes both gradient estimates and the paths traversed through the objective. Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments reveal loss spikes associated with pre-rise gradient reversal and norm amplification. Loss-defined events quantify their magnitude and recovery, while descriptive regressions relate the initial regime and subsequent norm growth to relative peak loss. The resulting framework supports the formulation and evaluation of informed learning-rate interventions; it does not prescribe an alignment-maximizing schedule.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Learning-Rate Scheduling**

I. Introduction

Neural scaling laws have ushered in new paradigms for interpreting deep learning dynamics. By establishing that performance gains follow predictable behavior relative to model size, dataset scale, and compute, stochastic optimization methods have been pushed toward increasingly extended training horizons while operating under tight efficiency constraints [1]–[4]. In this regime, optimization stability becomes an important practical requirement, since unstable training may result in slow convergence and high sensitivity to hyperparameter choices. In practice, stability is often maintained through a set of heuristics, such as momentum, batch-size scaling, normalization, and learning-rate schedules [5]–[9]. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

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Classical local analyses relate the permissible step size to curvature in sharp directions of the loss [10], [11]. Stochastic extensions, including the break-even point framework, also account for curvature variation induced by sampling [12]. The Edge-of-Stability literature shows that deep networks can continue to make progress while oscillating near, or beyond, the nominal sharpness threshold [13]–[15]. These results motivate a complementary diagnostic question: how do the magnitudes and directions of successive gradients combine into parameter motion along the observed training trajectory? Gradient information already computed during optimization makes this question accessible without estimating the full Hessian spectrum or gradient-noise covariance at every update.

To examine these interactions, we describe parameter motion through gradient magnitudes and directional alignment. Starting from an untrained reference, training changes the parameters through which the model expresses what it has learned. Displacement provides a geometric description of that change; it does not quantify retained knowledge or establish whether the change improves the objective. Positive and negative gradient interactions respectively reinforce and cancel components of the accumulated motion. Their effect must be evaluated at the relevant temporal horizon and distinguished from loss reduction.

We develop an exact finite-time decomposition of SGD displacement that retains all cross-step interactions. Consecutive-gradient cosine supplies a local online diagnostic, while normalized displacement measures the fraction of accumulated step length retained as net motion. Local expansions relate gradient changes to directional curvature and sampling noise. The distinction between these quantities is consequential: a trajectory can reverse successive gradient directions while continuing to reduce loss, and sampling can change both the measured alignment and the subsequent path.

The same distinction becomes essential under adaptive optimization. Adam already computes raw gradients and moment estimates [16]; its parameter updates combine temporal averaging, coordinate scaling and the learning rate. Monitoring these successive quantities can reveal changes that a single alignment statistic would obscure. Raw-gradient diagnostics remain informative about the traversed loss landscape even though Adam transforms those gradients before updating the parameters. Controlled full-batch trajectories reveal repeated loss excursions associated with gradient reversal and norm amplification. We examine individual spikes, their population anatomy and the association between the pre-rise regime, subsequent norm growth and peak magnitude. These observations motivate interventions while distinguishing descriptive

evidence from a causal mechanism or a validated prediction rule.

Our contributions are:

- A displacement-based account of local and cumulative gradient interaction, with explicit connections to curvature, sampling noise and finite-step loss changes;
- Controlled full-batch and mini-batch experiments that distinguish directional reversal, displacement retention and objective progress;
- A practical extension to Adam that separates raw-gradient information, momentum and adaptive directions, supported by controlled training and an exploratory analysis of loss-spike anatomy across models and rates.

The remainder of the paper is organized as follows. Section II reviews training-stability perspectives. Section III develops the displacement formulation and its diagnostics. Section IV examines full-batch GD and mini-batch sampling. Section V extends the interpretation to Adam and outlines the empirical requirements for informed scheduling. Section VI concludes.

II. Stability Analysis of SGD

Intuitively, smooth optimization occurs when updates reinforce each other and gradually shift parameters toward a minimum of the loss. Larger steps can overshoot, oscillate, and partially undo preceding parameter changes. Understanding when such behavior remains productive, and when it produces instability, requires examining how successive updates interact. Two factors are central to this analysis: the *local curvature* of the loss landscape and the *stochasticity* introduced by mini-batch sampling.

A. Local Curvature of the Loss Function and Stability

Training is a discrete-time dynamical system whose local response depends on curvature. A step that produces smooth descent in a shallow direction can overshoot in a sharper one. The Hessian quantifies this directional variation. Classical analyses [10], [11] connect its eigenvalues to the learning rate and local stability.

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t and g_t a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, plain SGD follows

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$. With $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of its update map is

$$J_t = I - \eta H_t. \tag{2}$$

On a fixed quadratic objective, a perturbation along a Hessian eigendirection is multiplied by $1 - \eta\lambda_i$ at each step. For a positive-definite Hessian, all components contract precisely when

$$\max_i |1 - \eta\lambda_i| < 1 \quad \Longleftrightarrow \quad \eta\lambda_{\max}(H_t) < 2. \tag{3}$$

For $0 < \eta\lambda_i < 1$, contraction preserves the component's sign. For $1 < \eta\lambda_i < 2$, its sign alternates while its magnitude shrinks. Above two, the component alternates and grows. These cases distinguish smooth approach, convergent oscillation, and instability of the fixed quadratic dynamics.

Neural-network Hessians are strongly anisotropic: a few outlier directions can be much sharper than the spectral bulk [17], [18]. Monitoring $\lambda_{\max}(H_t)$ therefore provides valuable stability information, but doing so throughout training is expensive. Even matrix-free eigensolvers require repeated Hessian–vector products beyond the ordinary gradient computation. An online diagnostic that recovers curvature information from those existing gradients would make this geometric analysis accessible during routine training.

B. Mini-Batch Stochasticity

Mini-batch SGD adds a sampling-dependent component to the full-gradient direction [19]–[21]. For independent uniform batches of b examples sampled without replacement from $N > 1$ examples,

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &= \frac{N-b}{b(N-1)}\Sigma(\theta_{t-1}). \end{aligned} \tag{4}$$

Here Σ is the covariance of per-example gradients. Writing $h_i := \nabla\ell_i(\theta_{t-1})$ and $\mu := N^{-1}\sum_i h_i = \nabla\mathcal{L}(\theta_{t-1})$,

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}(h_i - \mu)(h_i - \mu)^\top. \tag{5}$$

Its trace measures gradient variability, while its directional structure describes how sampling perturbs the trajectory. Exact evaluation requires per-example information absent from the aggregated training gradient. For an inexpensive online diagnostic, the practical question is how this sampling variability changes the alignment of the gradients already being computed.

C. Curvature, Noise and the Edge of Stability

Learning rate and batch size also shape the geometry encountered during training. The break-even framework of Jastrzebski et al. [12] shows how their early values influence later curvature and gradient variability. This motivates examining curvature and sampling together along the trajectory.

The Edge-of-Stability (EoS) phenomenon makes the distinction between local instability and sustained convergence especially clear. Full-batch GD often sharpens progressively until $\lambda_{\max}(H_t)$ fluctuates near or above $2/\eta$, while loss continues to decrease over longer timescales [13], [14]. A fixed quadratic with the same sharpness would be unstable, but a neural network changes its curvature as it moves. Self-stabilization analyses explain how growth along a sharp direction can induce a reduction in curvature, limiting the oscillation [15]. More recently, central flows describe the time-averaged trajectory together with the covariance of its rapid oscillations [22].

These results explain how short-term reversals can accompany longer-term progress. They also make the interaction

between successive updates central to understanding convergence: the trajectory can repeatedly reverse in sharp directions while continuing to descend elsewhere.

D. Practical Stability Analysis in Large-Model Pretraining

Practitioners address spikes through monitoring, recovery, and preventive changes to the training recipe. PaLM recovered from spikes by returning to an earlier checkpoint and skipping batches [23]. OPT combined checkpoint recovery with learning-rate reduction and a tighter gradient-clipping threshold [24]. These interventions act on both the magnitude of the update and the state from which training continues.

Recent open-weight reports also emphasize prevention. OLMo 2 tracks spikes in loss and gradient norm while evaluating initialization, normalization, and optimizer choices [25]. Kimi K2's MuonClip optimizer uses attention-logit-based rescaling of query and key weights and reports training without loss spikes [26]. Llama 3 reports few spikes and no divergence interventions, while DeepSeek-V3 reports no irrecoverable spikes or checkpoint rollbacks [27], [28].

Gradient norms are thus already useful monitoring signals. Their relationship to stability becomes clearer when magnitude is combined with direction: a growing gradient can reinforce or oppose its predecessor. We develop this connection through constructive and destructive regimes, showing how consecutive gradients estimate directional curvature and reveal the conditions for a local loss increase. This provides a geometric interpretation of signals available during training, using one retained gradient and no Hessian computation.

III. Constructive and Destructive Regimes

Loss spikes and oscillatory updates motivate diagnostics that explain how training is progressing as it happens. The gradients already computed by the optimizer contain two complementary signals: how strongly the loss changes locally, and whether the next update reinforces or opposes the preceding one. We first show how these interactions determine accumulated parameter change. We then connect consecutive alignment to curvature, obtaining an inexpensive estimator and a joint condition on gradient reversal and growth that explains when the loss can increase.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{6}$$

For training from scratch, θ_0 is the initial model and θ_T contains the parameter changes accumulated while learning from the data. Expanding their squared difference separates individual step magnitudes from interactions across steps:

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 &\sum_{t=1}^{T} \|g_t\|^2 \\ &+ 2\eta^2 \sum_{1\le u<t\le T} g_u^\top g_t.\end{aligned} \tag{7}$$

The squared norms are nonnegative contributions from individual steps. In full-batch GD, the negative gradient is the direction of steepest local descent, and a larger norm means a steeper local slope. The cross-step terms determine how these changes accumulate. Positive inner products reinforce displacement; negative inner products produce *destructive interference*, with one update undoing a component of another. Large individual steps can therefore coexist with limited net parameter change.

To make the respective roles of magnitude and direction explicit, factor each inner product as $g_u^\top g_t = \|g_u\|\|g_t\|\rho_{u,t}$, where

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \qquad 1 \le u < t \le T. \tag{8}$$

For nonzero consecutive gradients, write $\rho_t := \rho_{t-1,t}$. Substitution into Eq. (7) gives

$$\begin{aligned}\|\Delta_T\|^2 = \eta^2 &\sum_{t=1}^{T} \|g_t\|^2 \\ &+ 2\eta^2 \sum_{1\le u<t\le T} \|g_u\|\|g_t\|\rho_{u,t}.\end{aligned} \tag{9}$$

Constructive and destructive behavior are thus encoded directly in the SGD update equations. For fixed norms, positive alignment increases net displacement, whereas negative alignment limits it through cancellation. We identify a *constructive regime* by persistent $\rho_t > 0$ and a *destructive regime* by persistent $\rho_t < 0$. For plain SGD, these are also the cosines between successive parameter updates. They describe how learning changes the parameters; the associated loss determines whether those changes improve the objective.

A. Directional Curvature and Consecutive Alignment

We now connect these regimes to the curvature encountered along a step. For mini-batch SGD, use $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$ from Eq. (4). Suppose $\mathcal{L}$ is twice continuously differentiable along the preceding step and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} gives

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{10}$$

where r_t is the Taylor remainder. Neglecting it and projecting onto the preceding gradient yields

$$\begin{aligned}\|g_{t-1}\|\|g_t\|\rho_t \approx\ & \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ & \underbrace{+\, g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}.\end{aligned} \tag{11}$$

Define the curvature along the preceding gradient and the normalized sampling contribution as

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{12}$$

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

Dividing Eq. (11) by the two gradient norms gives the central relation

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{14}$$

In full-batch GD, $\varepsilon_t = 0$ and the norm ratio is positive. Within this approximation, $\rho_t \propto 1 - \eta\kappa_t$: $\eta\kappa_t < 1$ gives constructive alignment, while $\eta\kappa_t > 1$ gives destructive alignment. The boundary $\eta\kappa_t = 1$ separates the two regimes. At this boundary the gradients are orthogonal if both are nonzero; a vanishing gradient leaves the cosine undefined.

This relation supports two interpretations. From a representation-learning perspective, a destructive update opposes parameter changes made by its predecessor, providing a mechanism for repeatedly overwriting learned structure. Gradient conflicts also arise in catastrophic forgetting: continual-learning methods constrain updates against gradients of earlier tasks to preserve their losses [29]. Establishing forgetting in our setting would additionally require measuring retention of learned behavior. From an optimization perspective, negative alignment indicates that a GD step has passed the minimum along its preceding direction in the local quadratic model. The optimizer then partially reverses that step. Such reversals can coexist with convergence; the magnitude of the returning gradient determines when they instead increase the loss.

B. Estimating Curvature without Hessian Computation

Equation (14) turns consecutive training gradients into a directional-curvature estimator:

$$\begin{aligned} \eta\widehat{\kappa}_t &:= 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \\ &= -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2} \approx \eta\kappa_t - \varepsilon_t. \end{aligned} \tag{15}$$

Its computation requires retaining one additional gradient and taking norms and an inner product: $O(p)$ storage and arithmetic for p parameters, with no additional backward pass, Hessian–vector product, or eigensolver. Thus curvature information can be monitored at every training step using gradients the optimizer already needs.

The measured projected change has a precise geometric meaning. In smooth full-batch GD, $\widehat{\kappa}_t$ is exactly the directional Hessian curvature averaged along the preceding step; it approximates the starting-point Rayleigh quotient κ_t when curvature varies little over that step. Appendix A gives the identity and an error bound. It measures curvature along the realized gradient direction, while $\lambda_{\max}(H_t)$ measures the sharpest direction. In SGD, the additional term $-\varepsilon_t$ incorporates sampling. Larger batches reduce the gradient sampling variance and can bring the diagnostic closer to its full-batch interpretation. Its importance still depends on gradient variability relative to the full gradient, which we examine next.

C. Noise Contribution under Mini-Batch Sampling

Fresh, conditionally unbiased batches satisfy $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. Consequently, the noise contribution to consecutive alignment has nonpositive expectation:

$$\mathbb{E}[g_{t-1}^\top (\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{16}$$

For uniform batches sampled without replacement,

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)} \mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{17}$$

Smaller batches therefore increase the magnitude of this negative unnormalized contribution at fixed parameters; full-batch evaluation eliminates it.

Cosine attenuation also depends on normalization: sampling increases gradient variability relative to the direction shared between batches. For two independent gradients sampled at the same fixed state θ, concentration of their norms gives the approximation

$$\mathbb{E}[\rho_t \mid \theta \text{ fixed}] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)} \operatorname{tr}\Sigma(\theta)}. \tag{18}$$

Reducing b weakens this shared directional signal, moving the expected cosine toward zero. Along an evolving SGD trajectory, sampling also changes the next parameter state and its curvature. The negative term in Eq. (16) alone therefore does not determine how a negative cosine changes. Section IV-D tests the attenuation of constructive and destructive alignment on actual training trajectories.

D. Normalized Displacement and Cumulative Interference

How much of the distance traveled during training remains as net parameter change? We measure this fraction by the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta \sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{19}$$

defined for nonzero total path length. For constant-step plain SGD, the denominator is the traveled distance and the numerator is displacement from initialization. Equation (9) shows that R_T^2 incorporates the interactions between every pair of steps. Thus R_T summarizes the cumulative effect of their magnitudes and alignment, while ρ_t identifies the local regime.

Aligned steps give $R_T = 1$. Equal opposing pairs give zero, and equal-length orthogonal steps give $R_T = 1/\sqrt{T}$. Even without direct cancellation, orthogonal steps retain only 10% of their traveled distance after 100 updates, or 1% after 10,000. The example shows how weak directional reinforcement limits net movement relative to path length. Low R_T therefore reveals repeated reversal or weak coherence; its effect on learning is assessed jointly with the loss.

E. Conditions for Loss Decrease and Spike Formation

When does a destructive regime produce a loss increase? Define the full-objective change over the preceding step as $\Delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned} \Delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t \|g_{t-1}\| \|g_t\|}_{\text{Directional interference}} \\ & \underbrace{+\frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t)}_{\text{Noise interaction}}. \end{aligned} \tag{20}$$

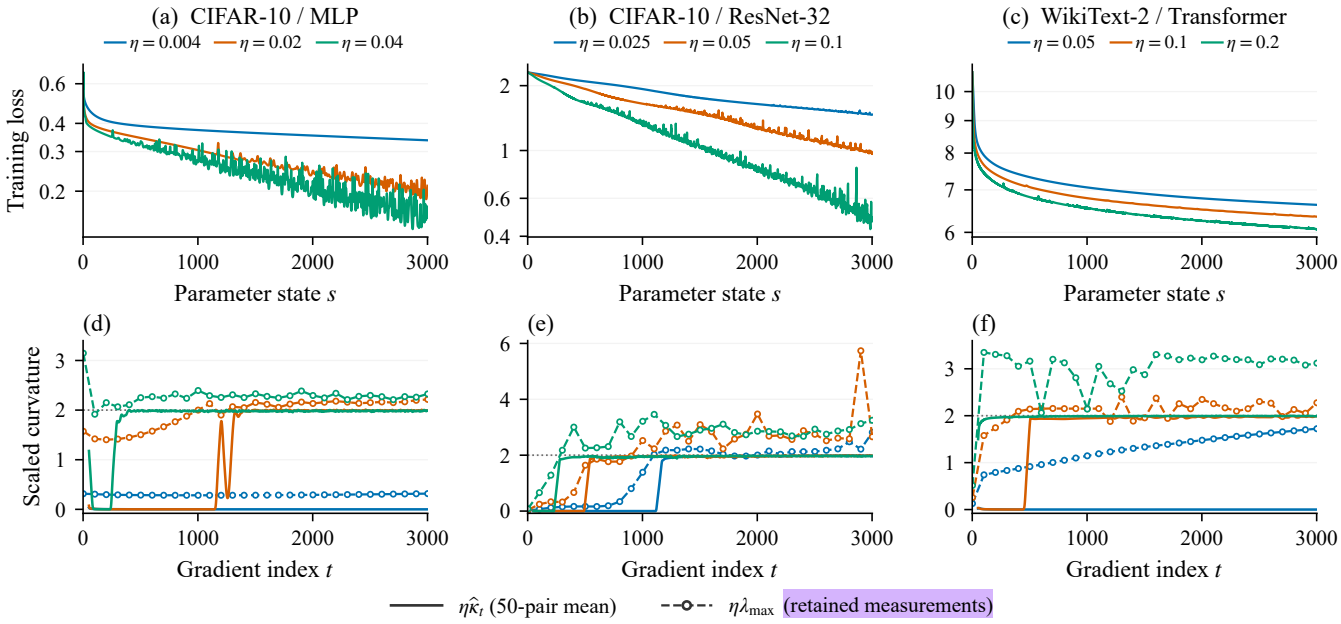


Fig. 1: Training loss and curvature diagnostics under full-batch GD. Columns identify dataset/model; colors identify learning rate. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: $\eta\widehat{\kappa}_t$ (solid, complete 50-pair trailing means) and retained $\eta\lambda_{\max}$ measurements (dashed with circles); the dotted line marks two. Spectra use the complete MLP objective, 5,000/50,000 ResNet images, and 2,500/59,680 Transformer sequences. Loss uses parameter state s; a spectrum at state s is placed at gradient index $t = s + 2$, the starting state for $\widehat{\kappa}_t$. The final spectrum has no recorded gradient pair.

The first term favors descent. Constructive alignment adds a further decrease, whereas destructive alignment offsets part of it. The size of this opposing contribution depends jointly on reversal and gradient growth. In full-batch GD with nonzero gradients, the approximation predicts $\Delta\mathcal{L}_{t-1} > 0$ precisely when

$$\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|} \iff g_{t-1}^{\top} g_t < -\|g_{t-1}\|^2. \tag{21}$$

Two conditions are necessary: $\rho_t < 0$ and $\|g_t\| > \|g_{t-1}\|$. The new gradient must reverse direction strongly enough that its opposing component exceeds the preceding magnitude. For example, doubling the norm requires $\rho_t < -0.5$, an angle above $120°$. A fivefold increase requires only $\rho_t < -0.2$, or about $101.5°$. Greater norm growth therefore makes even a milder reversal sufficient for an upward loss step.

The equivalent curvature form is

$$\Delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{22}$$

It summarizes the same joint effect: $\eta\kappa_t = 1$ marks the constructive–destructive boundary, while $\eta\kappa_t = 2$ marks the loss-increase boundary. Between them, destructive updates still decrease the objective. The expressions are exact for quadratic objectives; for nonlinear losses their accuracy depends on variation along the step, with sampling adding the explicit noise contribution. A sequence of upward steps followed by recovery forms the spike events studied below. The diagnostic thus connects local curvature to the reversal and norm growth underlying their formation.

IV. Empirical Characterization of Constructive and Destructive Regimes

We test how constructive and destructive regimes emerge during training, how the gradient-based curvature estimate relates to their onset, and how repeated reversal shapes accumulated parameter change. Full-batch GD isolates these dynamics from sampling noise. We then vary batch size to examine how stochasticity changes the observed alignment.

The comparison uses a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its fixed training objective for 3,000 updates. The constant learning rates are $\{0.004, 0.02, 0.04\}$ for the MLP, $\{0.025, 0.05, 0.1\}$ for ResNet, and $\{0.05, 0.1, 0.2\}$ for the Transformer. Within each model, all rates share one frozen initialization and data preparation. Prior training calibrations selected contrasting regimes; test loss did not guide selection. Plain GD uses no momentum, weight decay, clipping, or schedule. These three cases test recurrence across objectives and architectures, conditional on their matched initializations. Appendix B specifies the objectives and measurement conventions.

A. Curvature and Oscillatory Convergence

Figure 1 shows a clear separation between smooth and oscillatory convergence. The lowest MLP and Transformer

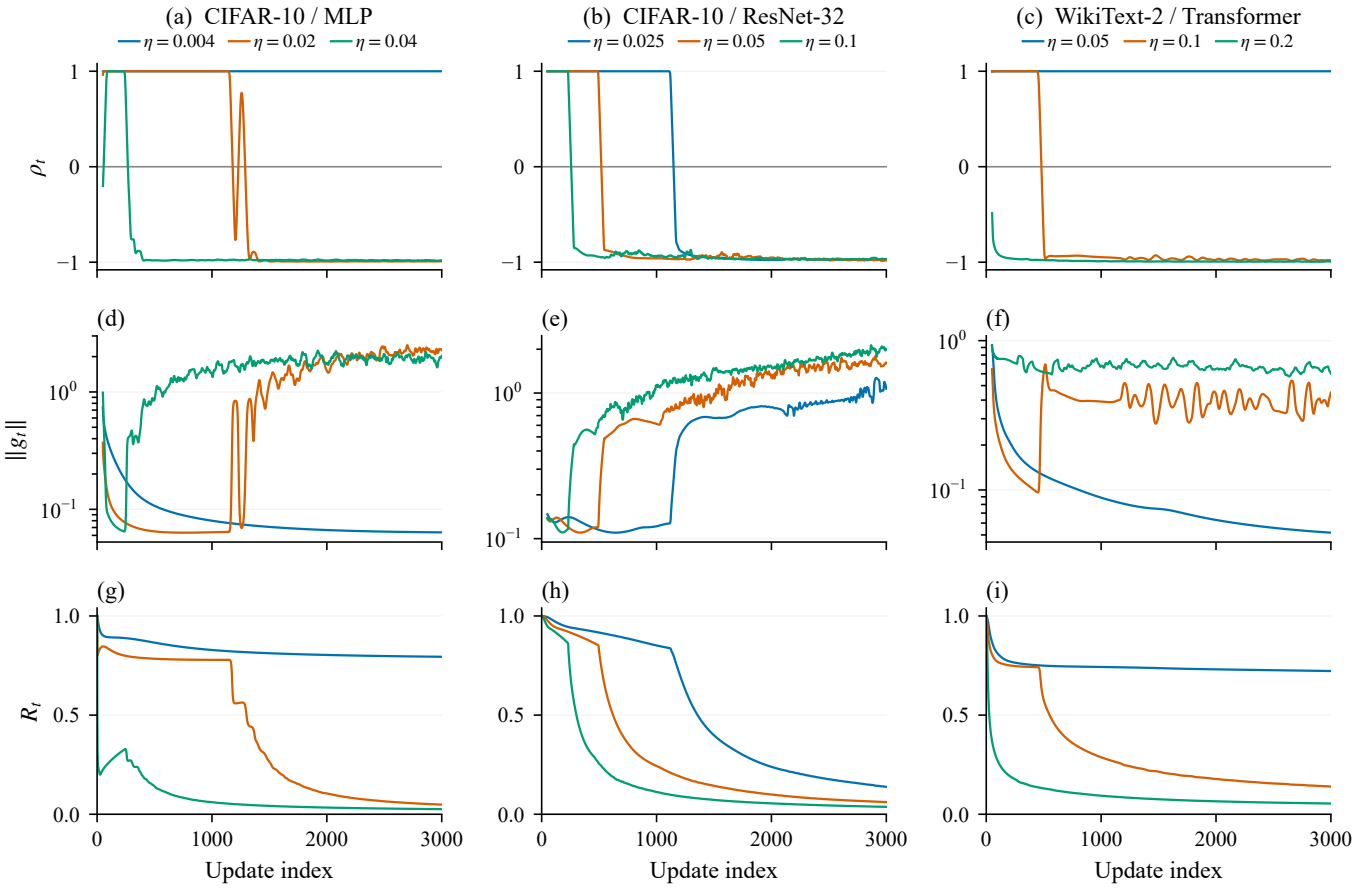


Fig. 2: Constructive/destructive alignment and cumulative motion for the conditions in Fig. 1. Rows show consecutive cosine, gradient norm (logarithmic scale), and normalized displacement. Cosines and norms use complete 50-observation trailing means; R_t is the unsmoothed ratio over the prefix through update t. Colors and column ordering match Fig. 1.

rates produce smooth loss curves and $\eta\widehat{\kappa}_t$ near zero. At higher rates, the loss develops fluctuations while its longer-term trend continues downward; the estimated scaled curvature approaches two. ResNet enters this regime at every tested rate, with earlier transitions at larger learning rates.

The retained Hessian measurements place these signatures in their spectral context. As sharpness grows, the trajectories develop the oscillations associated with EoS, while the inexpensive gradient diagnostic identifies the curvature response along the direction actually followed. In the oscillatory regime, $\eta\widehat{\kappa}_t$ remains close to the local loss-change boundary even when $\eta\lambda_{\max}$ is larger. The distinction is expected: directional curvature depends on the gradient's orientation, whereas the leading eigenvalue describes the sharpest direction. ResNet and Transformer spectra additionally use fixed subsets of their training objectives. The comparison therefore supports monitoring the realized curvature response throughout training, with no additional Hessian computation.

B. Destructive Regimes and Displacement Retention

The loss oscillations coincide with a transition to destructive alignment in Fig. 2. The lowest MLP and Transformer rates retain positive cosine, while the other seven conditions develop nearly opposing consecutive gradients. In the MLP and ResNet, this transition also accompanies growing gradient norms: larger steps increasingly reverse preceding parameter changes. The joint behavior directly exposes the magnitude and directional factors in Eq. (21).

Normalized displacement then declines strongly. The larger steps accumulate path length faster than net parameter change, so R_t falls even as training continues to reduce the loss. At the highest rates, the retained fraction after 3,000 updates is approximately 2.6% for the MLP, 3.7% for ResNet, and 5.4% for the Transformer. Nevertheless, each highest-rate condition reaches its model's lowest final training loss. Destructive regimes therefore sustain convergence despite extensive local reversal. The risk of an upward loss excursion depends on how strongly gradient growth amplifies that reversal.

C. Oscillation across Temporal Lags

We next ask whether destructive alignment reflects an isolated pair of updates or a recurring temporal structure. Figure 3 fixes the gradient after 2,256 updates and compares it with the preceding 256 gradients, using the highest rate for each model. The first lags alternate between strongly negative odd-lag and strongly positive even-lag alignment. This is direct evidence

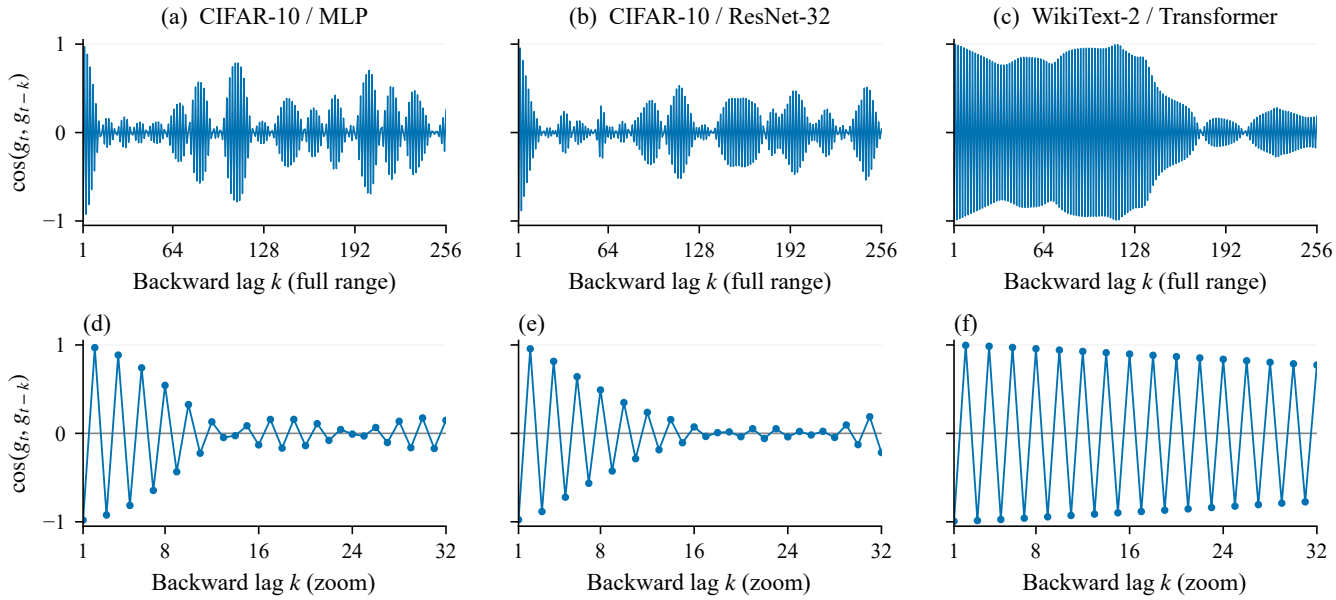


Fig. 3: Backward gradient alignment at the common endpoint θ_{2256}, corresponding to $t = 2257$. Each point is one exact $\cos(g_t, g_{t-k})$ with fixed endpoint. Top: lags 1–256. Bottom: the same values at lags 1–32. Rates are 0.04 (MLP), 0.1 (ResNet) and 0.2 (Transformer). These separately recomputed 256-update GD continuations start from retained state 2000, with one initialization, no smoothing and no endpoint averaging.

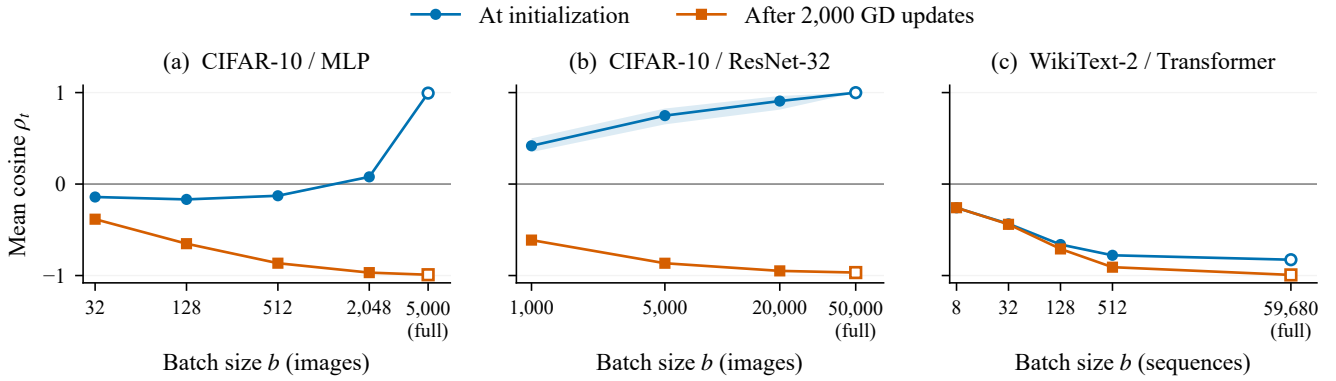


Fig. 4: Mean sampled-gradient cosine over 199 adjacent pairs during 200-update continuations. Blue circles start at initialization; orange squares start after 2,000 GD updates. Each line averages repetitions within initialization, then weights the three initialization blocks equally; shading is their range. Hollow rightmost markers denote full GD. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic.

of repeated back-and-forth motion: a gradient can oppose its immediate predecessor while reinforcing the direction from two steps earlier.

The early oscillation envelope contracts, rapidly for the MLP and ResNet and more slowly for the Transformer. Across the full window, alignment can strengthen again, so the envelope is not monotone. These recurrences explain how local destructive interference coexists with reinforcement over longer lags and why net progress survives repeated reversals.

D. Mini-Batch Attenuation of Directional Alignment

To test the effect of sampling, we vary batch size at initialization and after 2,000 GD updates in the same three model–dataset cases. Learning rates remain fixed at 0.02, 0.1, and 0.2 for the MLP, ResNet, and Transformer, respectively. Each starting state receives 200-update continuations, with three initializations, eight sampling repetitions per stochastic condition, and one full-GD control per initialization. Figure 4 summarizes mean consecutive-gradient cosine over these continuations. Sampling units and objective coverage are specified in Appendix B; equal update budgets imply different amounts of data exposure.

From the trained states, the smallest batches move the strongly negative full-GD alignment toward zero in all three cases. This attenuation is consistent with sampling weakening the coherent back-and-forth motion of the destructive regime.

The response need not be monotone across intermediate batch sizes: sampling changes both the measured gradient and the trajectory itself.

The initial-state comparison makes this dependence on the trajectory visible. Full GD has positive mean cosine in the MLP and ResNet, but negative mean cosine in the Transformer, where reversal develops within the 200-update window. Smaller MLP batches can also induce negative mean alignment despite a positive full-GD mean. Thus the batch size influences which regime training reaches as well as the coherence within that regime.

Directional attenuation alone does not determine objective progress. For the Transformer at $b = 8$, mean cosine is near -0.26 from both starting states, yet mean full-data loss falls from initialization and rises from the trained state. Together with the full-batch results, this motivates reading alignment alongside gradient magnitude and loss: the same directional regime can support descent or an upward excursion depending on the gradient's growth and the state being traversed.

V. Raw-Gradient Diagnostics under Adam

Adam [16] transforms the raw gradient through first- and second-moment estimates,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{23}$$

with zero initial moments. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{24}$$

where operations are coordinatewise. Although d_t determines the step, the raw gradients still reveal variation in local slope along the adaptive trajectory. Their consecutive alignment and magnitudes distinguish directional reversal from a sharp increase in local slope. The first moment shows how this changing input propagates through the optimizer's memory.

The projected-change diagnostic remains exactly computable as

$$\eta\widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|}. \tag{25}$$

Under Adam, this is normalized projected raw-gradient change along the realized path; it is not the SGD directional-curvature estimate. A value above two requires negative cosine and a sufficiently large consecutive norm ratio. Thus, reversal and norm growth explain the diagnostic's increase algebraically, rather than supplying independent confirmations of instability. The reference at two recalls the GD analysis, but does not identify an Adam stability boundary.

A. Controlled Full-Batch Training

We reuse the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each Adam trajectory starts from the same frozen, untrained seed-0 weights used for its GD comparison, with fresh moments. We run 1,024 full-objective updates at each constant rate: $\{0.0001, 0.0003, 0.001\}$ for the MLP and $\{0.0003, 0.001, 0.003\}$ for ResNet and Transformer. All nine arms use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$, without a schedule, clipping or weight decay. There is one initialization per model, matched across rates. Full-batch accumulation removes sampling variation in the objective; model, objective and numerical details are in Appendix B-C.

Figure 5 shows overall convergence interspersed with transient loss spikes, especially at larger rates. The largest tested rate attains the lowest final training loss in each model despite these excursions. A spike can therefore interrupt a trajectory that continues to improve; its presence alone does not rank the final loss. We next examine the gradient dynamics within individual excursions before summarizing their distribution across the nine arms.

B. Anatomy of Individual Spikes

For each model, we select the recovered excursion with the largest peak/pre-rise loss ratio at its highest tested rate. Selection uses only loss, with no condition on a diagnostic. The selected peaks occur at states 703, 422 and 794 for MLP, ResNet and Transformer, respectively. Their location in the middle or later part of training is an outcome of this maximum-amplitude rule, not an additional temporal filter.

All three examples in Fig. 6 enter the pronounced rise with negative raw-gradient cosine. During the rise, the gradient norm increases sharply and $\eta\widehat{\kappa}$ exceeds two before the loss maximum. Momentum norms also grow, with model-dependent timing relative to the raw gradient. In the Transformer, the first crossing of two within the excursion is at state 788, when loss is only 0.40% above its pre-rise level. The +10% crossing follows at 790; raw-gradient and momentum norms reach their event maxima at 792 and 793, before loss peaks at 794. The diagnostic thus precedes the pronounced spike and its maximum, while the first small loss increase has already occurred.

The descent separates the models. ResNet and Transformer move toward positive gradient and momentum cosines, lower projected change and a comparatively smooth loss decline. The MLP continues to oscillate: raw cosine remains mostly negative while momentum cosine changes sign, and loss and norms decrease through repeated rebounds. Recovery therefore need not require every raw-gradient pair to become constructive.

These examples associate the developing loss spike with gradient reversal and rapid norm amplification, even though the parameter steps are adaptive. An above-two value marks a strong local gradient response consistent with the observed excursion; it is not proof of a linear instability of Adam. The early response suggests an opportunity to anticipate further growth and intervene. Whether it supplies a reliable trigger, or whether acting on it improves training, requires evidence beyond selected examples.

C. Loss-Defined Events and Their Distribution

To examine repetition without selecting on gradients, write $\mathcal{L}_s := \mathcal{L}(\theta_s)$ for the recorded training loss. Starting at initialization, let a_i be the earliest available state with $\mathcal{L}_{a_i+1} > \mathcal{L}_{a_i}$. Hold the pre-rise reference $\mathcal{L}_{a_i}$ fixed. When a return is

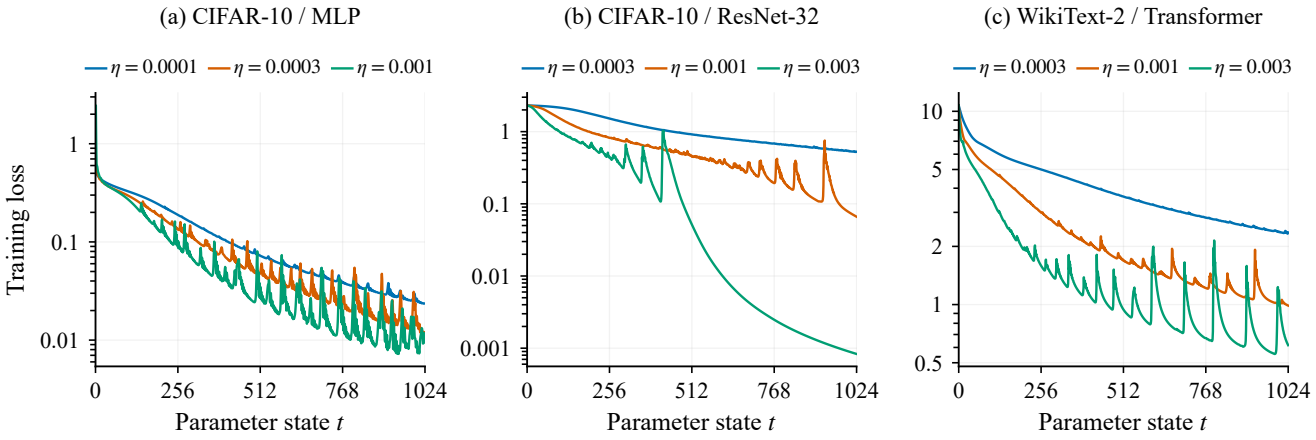


Fig. 5: Full-batch Adam training from matched initial weights, with three constant learning rates per model. Every training-loss observation is shown without smoothing on logarithmic y-axes. Parameter state s denotes s completed updates, from initialization to 1,024 updates. One initialization per model; colors identify increasing rates within each panel.

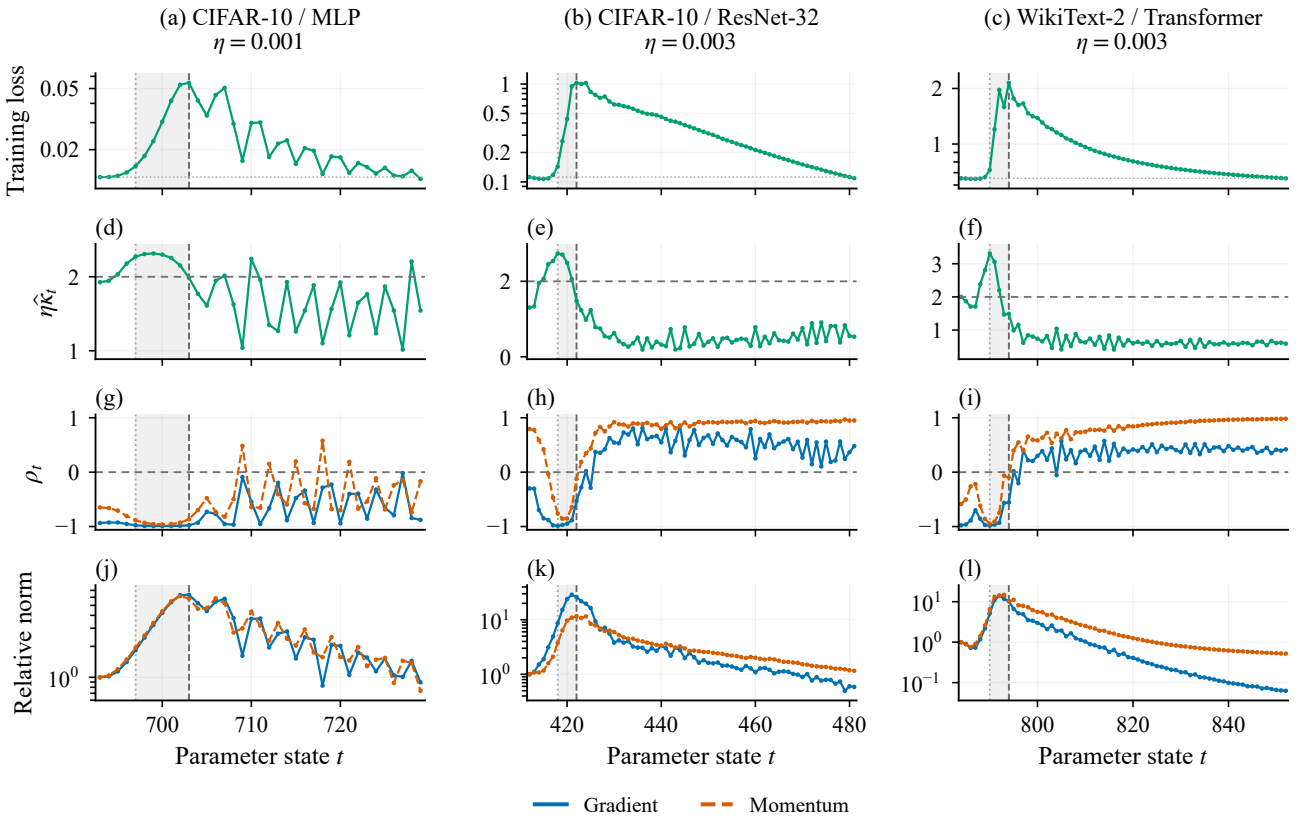


Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show loss, raw-gradient projected change, consecutive cosines of raw gradients and first moments, and their relative norms. All observations are unsmoothed. At parameter state s, vector diagnostics refer to g_{s+1} and m_{s+1}. Norms are divided by their value at the first displayed state. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak (697–703, 418–422 and 790–794), rather than the entire excursion through recovery. The horizontal dotted loss reference is the first displayed loss. Windows begin ten states before the peak and end at the first later loss below that displayed reference.

observed within T updates, define the first recovery and peak by

$$c_i := \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\},$$
$$p_i := \min \arg\max_{a_i < s < c_i} \mathcal{L}_s. \tag{26}$$

An event is included only when recovery is observed by $T = 1024$. Search resumes at c_i; partial declines and additional local maxima before this return belong to the same event. Thus a_i is the last pre-rise state, $a_i + 1$ the first elevated state, p_i the loss peak, and c_i the first recovered state. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$, in optimizer updates. The three loss ratios are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

Figure 7 places the illustrations within the complete distribution of recovered excursions. Small fluctuations coexist with long tails; larger rates populate more severe excursions, although the distributions overlap and do not undergo a uniform shift. To focus the anatomy analysis on pronounced changes, we retain events with peak increase strictly greater than 10%. This is an operational amplitude filter chosen for interpretation, not a separation inferred from a density minimum or an optimizer stability threshold. The density retains all amplitudes so that the selection remains visible in context. The filtered sample contains 129 recovered events. Two further above-threshold candidates without observed recovery are excluded.

For Table I, pre-rise diagnostics are measured at θ_{a_i}, hence use gradient index $a_i + 1$. Recovery entries first average each event's diagnostic over the arrival states $p_i + 1, \dots, c_i$, then summarize those means across events with equal event weight. They are not values only at c_i. The norm-ratio columns instead use the endpoints $\|g_{p_i+1}\|/\|g_{a_i+1}\|$ and $\|g_{c_i+1}\|/\|g_{p_i+1}\|$. The gradient norm's own maximum need not coincide with the loss peak.

Among qualifying events, both the mean increase and the maximum increase grow with LR in every model with more than one eligible rate. The largest increase is 858.8% for ResNet at $\eta = 0.003$. Mean recovery takes 17–19 updates versus 6–8 for the ResNet rise, and 24–31 versus about 6 for the Transformer. MLP rise and recovery times are more comparable. Thus, a short amplification can be followed by a much longer descent, without implying that all intermediate loss changes are monotonic.

Pre-rise raw cosine is negative in 117 of 127 defined events, and the gradient norm at the loss peak exceeds its pre-rise value in all 129. Projected change is already above two before the first loss increase in 21 of 127 defined cases. Its full-event maximum precedes the loss peak in 82 of 129 events, coincides in 29 and follows in 18. These observations support recurrent sign-and-magnitude structure, rather than a universal pre-onset crossing. Recovery is also heterogeneous: event-averaged raw cosines remain negative for the MLP and the eligible ResNet groups, and are near zero for the Transformer. The positive recovery segments in the illustrations therefore do not describe every event.

D. Reversal, Norm Amplification and Possible Interventions

To relate the initial regime to amplification during an event, we plot

$$\rho_{a_i+1} \frac{\|g_{p_i+1}\|}{\|g_{a_i+1}\|} \quad \text{against} \quad \frac{\mathcal{L}_{p_i}}{\mathcal{L}_{a_i}}. \tag{27}$$

This combines the pre-rise consecutive cosine with the subsequent rise norm ratio. It is an event descriptor, not Eq. (25), which pairs a cosine with a *consecutive* norm ratio. Its magnitude incorporates the future peak and cannot be used unchanged as an online predictor.

Figure 8 relates larger spikes to the combination of a destructive initial gradient regime and norm amplification. The per-rate fits explain 67–69% of log-loss-ratio variation for the MLP, 90–94% for ResNet, and 38–54% for the Transformer. The association is consistent with the qualitative anatomy, with substantial remaining model-dependent dispersion. It does not establish a linear relation on the original loss scale, an Adam stability criterion or a causal feedback loop. Both axes use loss-defined endpoints, the sample is conditional on future amplitude and recovery, and events within a trajectory can be dependent.

This anatomy motivates testing whether intervention during reversal and early norm growth can attenuate the subsequent excursion. Gradient-norm clipping was proposed to address exploding gradients [30], and analyses allowing smoothness to grow with gradient norm provide a further rationale for adaptive scaling [31]. Our observations are consistent with investigating norm control, without identifying those theoretical conditions or establishing that clipping would help these Adam runs. Clipping Adam's input changes both moment streams; its effect on the adaptive step cannot be inferred from the raw norm alone. Learning-rate adjustment and data-mixing interventions provide other tests of the proposed interaction between the current regime and subsequent gradient growth. These tests should isolate states before pronounced excursions, compare matched interventions with unchanged continuations, and measure both spike attenuation and later loss at equal budgets. Establishing prospective warning information also requires non-spike comparisons. Such interventions and predictive validation remain future work.

VI. Conclusion

The displacement formulation distinguishes the magnitude of individual updates, their local directional interaction and their cumulative accumulation. For plain SGD, the exact decomposition retains all temporal lags, while consecutive cosine provides a tractable local observation. The curvature and sampling relations clarify how that observation can change without determining either cumulative displacement or loss improvement. The full-batch and mini-batch experiments illustrate these distinctions: gradient reversal can coexist with objective progress, and weaker reversal need not improve displacement retention or the attained loss.

Under Adam, raw-gradient diagnostics remain informative about the loss landscape traversed by adaptive steps. Controlled full-batch experiments connect loss spikes with a predominantly destructive pre-rise gradient regime and sharp

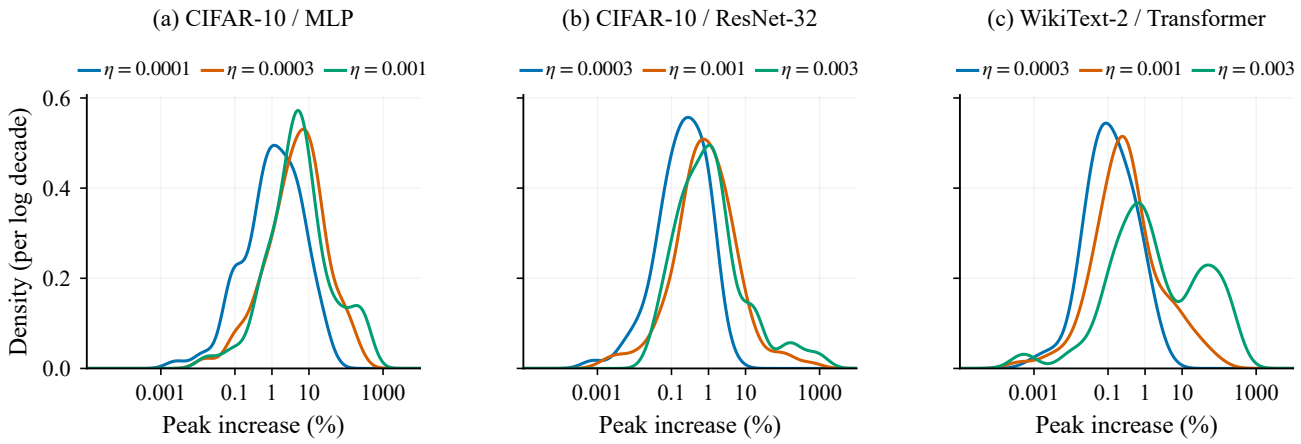


Fig. 7: Distributions of peak percentage increases for all 806 recovered excursions, including small fluctuations, with no amplitude cutoff. Colors identify LRs within each model. Gaussian kernel densities are estimated in $\log_{10}$ percentage-increase coordinates with one pooled bandwidth per model, common across its LRs. Each LR density integrates to one in that coordinate. The log x-axis shows percentage increase: 1 denotes +1%, and 100 denotes +100%. This differs from the loss-ratio axis in Fig. 8.

TABLE I: Anatomy of recovered spikes with peak increase greater than 10%. Cells report mean ± sample SD across events; maximum peak increase has its own subcolumn. Before denotes a single diagnostic at state a_i; recovery denotes the event's mean over states $p_i + 1, \ldots, c_i$. Norm ratios compare loss-peak/pre-rise and recovered/loss-peak endpoints. N counts included events; dashes indicate no qualifying event. Two initialization events have undefined pre-rise diagnostics, leaving 46 and 38 observations for those entries in the MLP's middle and highest LR rows. No smoothing or averaging across initializations.

Dataset Model	η	N	Peak increase (%)		Time (updates)		$\eta\widehat{\kappa}$ (raw)		ρ (raw)		Gradient norm ratio	
			Mean ± SD	Max	Rise	Recovery	Before	Recovery	Before	Recovery	Rise	Recovery
CIFAR-10 MLP	10^{-4}	8	19 ± 6.1	31	8.4 ± 2.9	4.4 ± 2.4	1.8 ± 0.38	1.7 ± 0.08	-0.77 ± 0.31	-0.89 ± 0.08	5.3 ± 1.7	0.30 ± 0.07
	3×10^{-4}	47	43 ± 46	203	5.5 ± 4.4	4.9 ± 4.0	1.3 ± 0.25	1.6 ± 0.18	-0.53 ± 0.29	-0.71 ± 0.24	4.5 ± 3.0	0.33 ± 0.13
	10^{-3}	39	95 ± 107	318	6.1 ± 5.9	7.4 ± 7.5	1.4 ± 0.40	1.4 ± 0.26	-0.46 ± 0.38	-0.55 ± 0.30	5.2 ± 3.8	0.31 ± 0.14
CIFAR-10 ResNet-32	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	10	105 ± 182	602	7.5 ± 2.2	17 ± 11	1.3 ± 0.39	1.3 ± 0.25	-0.36 ± 0.42	-0.36 ± 0.26	4.3 ± 2.9	0.43 ± 0.24
	3×10^{-3}	7	179 ± 309	859	6.3 ± 1.3	19 ± 20	1.7 ± 0.38	1.1 ± 0.37	-0.65 ± 0.28	-0.13 ± 0.42	5.0 ± 4.2	0.34 ± 0.24
WikiText-2 Transformer	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	6	33 ± 25	81	6.3 ± 1.9	24 ± 9.8	2.1 ± 0.27	0.96 ± 0.16	-0.84 ± 0.10	-0.02 ± 0.12	9.0 ± 3.8	0.09 ± 0.05
	3×10^{-3}	12	82 ± 70	231	6.2 ± 2.7	31 ± 17	2.2 ± 0.43	0.91 ± 0.19	-0.79 ± 0.22	0.09 ± 0.21	11 ± 6.9	0.08 ± 0.08

norm amplification. The micro-plots resolve projected gradient changes before the loss maximum; the event population quantifies the variation in timing and recovery across models and rates. A joint descriptor of pre-rise reversal and rise norm growth is associated with relative peak magnitude, without establishing a causal feedback mechanism or a universal instability threshold.

The practical contribution is a framework for interpreting these measurements together and using them to formulate controlled interventions. Developing a scheduler or clipping rule requires demonstrating that acting on early diagnostics improves subsequent loss at matched budgets, beyond decisions based on recent loss and known training settings. Prospective validation, controlled interventions and replication across independent runs remain future work.

Appendix A
Curvature Estimation and Local Loss Change

Write $H(\theta) = \nabla^2 \mathcal{L}(\theta)$ and assume the objective is twice continuously differentiable along the plain-SGD step. Integrate its gradient derivative on $\theta_{t-2} - s\eta g_{t-1}$, add the endpoint-noise difference, and project onto $-g_{t-1}/(\eta \|g_{t-1}\|^2)$. The result is

$$\widehat{\kappa}_t = \int_0^1 \frac{g_{t-1}^\top H(\theta_{t-2} - s\eta g_{t-1}) g_{t-1}}{\|g_{t-1}\|^2}\, ds - \frac{\varepsilon_t}{\eta}. \tag{28}$$

If H is L_H-Lipschitz in operator norm on the step, its full-batch specialization satisfies

$$|\widehat{\kappa}_t - \kappa_t| \le \int_0^1 L_H s\eta \|g_{t-1}\|\, ds = \frac{L_H \eta}{2} \|g_{t-1}\|. \tag{29}$$

This bounds the difference between the measured path average and the starting-point directional curvature. For Adam, inte-

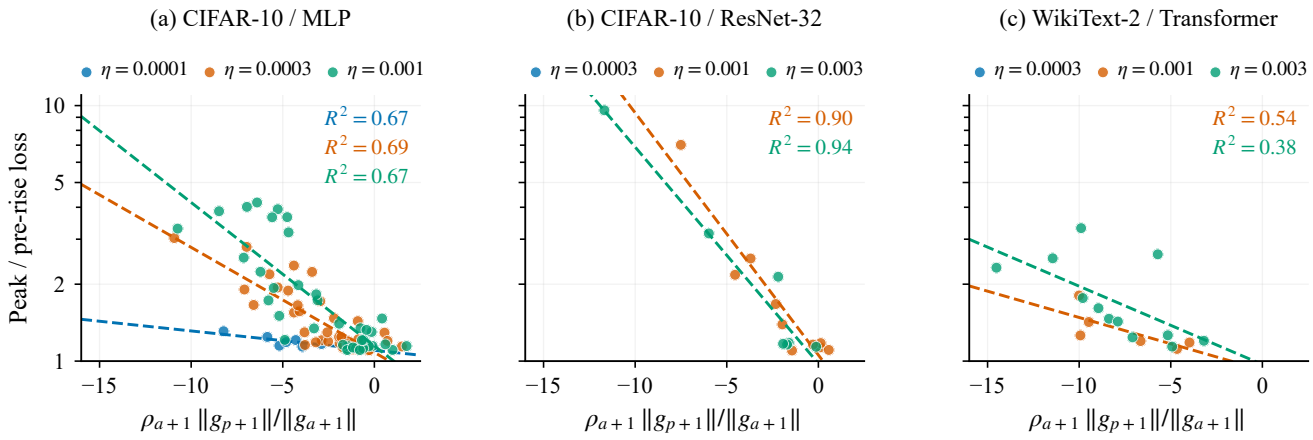


Fig. 8: Pre-rise raw-gradient cosine times rise norm ratio versus peak/pre-rise loss for 127 recovered spikes above 10%; two initialization events have undefined pre-rise cosine. One unsmoothed point per event. Colors identify LRs. Separate per-LR least-squares fits with intercept use $\log_{10}$(peak/pre-rise loss) as the response; color-matched R^2 values are descriptive in-sample scores on that log response. Dashed lines extend across each panel, including extrapolation beyond observed samples. The y-axis is a loss ratio: 1 denotes no increase and 2 denotes +100%.

grate along its adaptive step and project onto the preceding raw gradient. This gives the exact path-average form of Eq. (25).

Consider $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$ and write $\Delta\mathcal{L}_{t-1} = \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. Integrating the full gradient along the step gives

$$\Delta\mathcal{L}_{t-1} = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{30}$$

The trapezoidal approximation replaces the integrand by its endpoint average. Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $\nabla\mathcal{L}(\theta_{t-1}) = g_t - \xi_t$ yields

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ + \frac{\eta}{2} g_{t-1}^\top (\xi_{t-1} + \xi_t).\end{aligned} \tag{31}$$

The substitution $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ recovers Eq. (20). The expression is exact on quadratic objectives. A locally Lipschitz Hessian bounds its error by a constant times $\|\eta g_{t-1}\|^3$.

Using $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t + \varepsilon_t$ and Eq. (13) gives the equivalent curvature form,

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) \\ + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{32}$$

It also follows directly from a second-order Taylor expansion at θ_{t-2}. With zero noise, the conditions for a loss-increasing step reduce to Eqs. (21) and (22).

The noise term has a simple conditional interpretation. Given θ_{t-2}, unbiased sampling gives $\mathbb{E}[\|g_{t-1}\|^2] = \|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \mathbb{E}[\|\xi_{t-1}\|^2]$ and $\mathbb{E}[\xi_{t-1}^\top g_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]$. Their first-order variance terms cancel, leaving

$$\begin{aligned}\mathbb{E}[\Delta\mathcal{L}_{t-1} \mid \theta_{t-2}] \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{33}$$

Thus sampling contributes through curvature at second order. A nonlinear objective also contributes the finite-step remainder.

Appendix B
Essential Measurement Conventions

A. Fixed Objectives and Numerical Scope

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten outputs and averaged over the first 5,000 CIFAR-10 images. ResNet-32 minimizes mean cross-entropy over all 50,000 training images, using 50 fixed groups of 1,000 for batch normalization. Standardization uses the complete CIFAR-10 training split: coordinate-wise for the MLP and channel-wise for ResNet. Full-batch accumulation evaluates the same fixed objective at every update; augmentation and dropout are disabled.

The word-level Transformer has two layers, two heads and embedding/feed-forward dimensions of 200. Twenty corpus streams form causal chunks of at most 35 tokens, yielding 59,680 sequences and 2,088,600 supervised targets. The objective averages negative log-likelihood over valid targets, excluding padding. Architecture and the 33,278-word first-occurrence dictionary over training, validation and test text follow PyTorch example revision `13acec6d7c78` [32].

Diagnostics cover all trainable parameters. Parameters, gradients and Adam moments use FP32; reductions use FP64, with TF32 disabled. Initial unpaired cosines and zero-denominator ratios are undefined. The GD plots' trailing means summarize sustained regimes and can delay a displayed transition; Adam losses and micro-plots are unsmoothed. All claims concern training objectives. Matched tensors give one initialization per full-batch model; independent hardware executions can develop finite-precision drift.

The GD spectral comparison reuses largest algebraic Hessian eigenvalues at states $0, 100, \ldots, 3000$, with relative eigen-

pair residual at most 0.005. MLP spectra cover its complete 5,000-image objective; ResNet uses its first five fixed normalization groups, and the Transformer its first 2,500 sequences. These latter spectra characterize subset objectives. The earlier MLP solver requested two eigenvalues and the other runs one; the comparison consistently uses the largest.

B. Sampling and Trajectory Comparisons

Stochastic batches are independent uniform subsets without replacement; within a repetition, batch sizes share prefixes of a common permutation. ResNet samples complete normalization groups. Transformer sampling uses $N = 59{,}680$ sequences and $M = 2{,}088{,}600$ total targets: for sequence summed losses ℓ_i, the estimator is $N(bM)^{-1} \sum_{i \in B} \ell_i$. Its fixed denominator preserves the full token-weighted objective. The seed-2 smallest two ResNet batches use a different runtime, with 200-update full-GD replay parameter relative error 2.36×10^{-7}; batch and runtime remain confounded in this block. The backward-lag curves also use separate continuations, whose later floating-point paths can differ from the original GD trajectories.

C. Adam Event Summaries

At state θ_s, the raw gradient and updated first moment are g_{s+1} and m_{s+1}, before parameter update $s + 1$. The terminal raw gradient is evaluated without another update. Hardware varies across ResNet and Transformer learning-rate arms; initial-gradient agreement is checked, while subsequent finite-precision drift remains possible. Rate selection uses training calibration.

Two excursions above 10% without observed recovery are excluded. Two retained initialization events have undefined pre-rise diagnostics. Each event has equal weight; recovery diagnostics first average within the event, and sample SD uses denominator $n - 1$. The scatter uses per-rate ordinary least squares with an intercept on log relative peak loss. Its R^2 describes the fitting sample; dashed extensions extrapolate. For the density, each model's pooled log-percentage SD times pooled $n^{-1/5}$ sets the common Gaussian bandwidth. Events within a trajectory can be dependent.

REFERENCES

[1] J. Hestness, S. Narang, N. Ardalani, G. Diamos, H. Jun, H. Kianinejad, M. Patwary, Y. Yang, and Y. Zhou, "Deep learning scaling is predictable, empirically," *arXiv preprint arXiv:1712.00409*, 2017.

[2] J. Kaplan, S. McCandlish, T. Henighan *et al.*, "Scaling laws for neural language models," *arXiv preprint arXiv:2001.08361*, 2020.

[3] T. Henighan *et al.*, "Scaling laws for autoregressive generative modeling," *arXiv preprint arXiv:2010.14701*, 2020.

[4] J. Hoffmann, S. Borgeaud, A. Mensch, E. Cassirer, T. Jones, T. Dasilva, K. Tunyasuvunakool, C. Humphreys, E. Rutherford, E. Buchatskaya *et al.*, "Training compute-optimal large language models," *arXiv preprint arXiv:2203.15556*, 2022.

[5] P. Goyal, P. Dollár, R. Girshick, P. Noordhuis, L. Wesolowski, A. Kyrola, A. Tulloch, Y. Jia, and K. He, "Accurate, large minibatch sgd: Training imagenet in 1 hour," *arXiv preprint arXiv:1706.02677*, 2017.

[6] L. N. Smith, "Cyclical learning rates for training neural networks," in *2017 IEEE winter conference on applications of computer vision (WACV)*. IEEE, 2017, pp. 464–472.

[7] I. Loshchilov and F. Hutter, "Sgdr: Stochastic gradient descent with warm restarts," *arXiv preprint arXiv:1608.03983*, 2016.

[8] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International conference on machine learning*. pmlr, 2015, pp. 448–456.

[9] A. Krogh and J. Hertz, "A simple weight decay can improve generalization," *Advances in neural information processing systems*, vol. 4, 1991.

[10] B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics*, vol. 4, pp. 1–17, 1964.

[11] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Belmont, MA: Athena Scientific, 1999.

[12] S. Jastrzębski, M. Szymczak, S. Fort, D. Arpit, J. Tabor, K. Cho, and K. Geras, "The break-even point on optimization trajectories of deep neural networks," in *International Conference on Learning Representations (ICLR)*, 2020.

[13] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar, "Gradient descent on neural networks typically occurs at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2021.

[14] S. Arora, Z. Li, and A. Panigrahi, "Understanding gradient descent on the edge of stability in deep learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 948–1024.

[15] A. Damian, E. Nichani, and J. D. Lee, "Self-stabilization: The implicit bias of gradient descent at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2023.

[16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[17] L. Sagun, U. Evci, V. U. Guney, Y. Dauphin, and L. Bottou, "Empirical analysis of the hessian of over-parametrized neural networks," 2017.

[18] B. Ghorbani, S. Krishnan, and Y. Xiao, "An investigation into neural net optimization via hessian eigenvalue density," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2232–2241.

[19] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate bayesian inference," *Journal of Machine Learning Research*, vol. 18, no. 134, pp. 1–35, 2017. [Online]. Available: https://www.jmlr.org/papers/v18/17-214.html

[20] W. Hu, C. J. Li, L. Li, and J.-G. Liu, "On the diffusion approximation of nonconvex stochastic gradient descent," *Annals of Mathematical Sciences and Applications*, vol. 4, no. 1, pp. 3–32, 2019.

[21] S. Jastrzębski, Z. Kenton, D. Arpit, N. Ballas, A. Fischer, Y. Bengio, and A. Storkey, "Three factors influencing minima in sgd," *arXiv preprint arXiv:1711.04623*, 2017. [Online]. Available: https://arxiv.org/abs/1711.04623

[22] J. M. Cohen, A. Damian, A. Talwalkar, J. Z. Kolter, and J. D. Lee, "Understanding optimization in deep learning with central flows," in *International Conference on Learning Representations*, 2025.

[23] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts *et al.*, "PaLM: Scaling language modeling with pathways," *arXiv preprint arXiv:2204.02311*, 2022.

[24] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen *et al.*, "OPT: Open pre-trained transformer language models," *arXiv preprint arXiv:2205.01068*, 2022.

[25] Team OLMo, P. Walsh, L. Soldaini, D. Groeneveld, K. Lo *et al.*, "2 OLMo 2 furious," *arXiv preprint arXiv:2501.00656*, 2025.

[26] Kimi Team *et al.*, "Kimi K2: Open agentic intelligence," *arXiv preprint arXiv:2507.20534*, 2025.

[27] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[28] DeepSeek-AI *et al.*, "DeepSeek-V3 technical report," *arXiv preprint arXiv:2412.19437*, 2024.

[29] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[30] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *International conference on machine learning*. Pmlr, 2013, pp. 1310–1318.

[31] J. Zhang, T. He, S. Sra, and A. Jadbabaie, "Why gradient clipping accelerates training: A theoretical justification for adaptivity," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=BJgnXpVYwS

[32] PyTorch contributors, "PyTorch examples: Word language model," GitHub repository, revision `13acec6d7c78`; full commit specified in the URL. [Online]. Available: https://github.com/pytorch/examples/tree/13acec6d7c78dacd5e1fe9b0b4a325e1d39abc15/word_language_model